

Build a Virtual Private Cloud with VPC Wizard

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Purpose

I was shown ways to quickly make a VPC with a public and private subnet using VPC Wizard which has pre-sets like the ones I used in this lab. Additionally, I worked with Elastic IP addresses such as how to obtain and then release one, as well as looking at routing tables and connecting them to subnets.

Lab Walkthrough (with Documentation)

Step 1: Select a VPC Configuration

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VPC with a Single Public Subnet

VPC with Public and Private Subnets

VPC with Public and Private Subnets and Hardware VPN Access

VPC with a Private Subnet Only and Hardware VPN Access

In addition to containing a public subnet, this configuration adds a private subnet whose instances are not addressable from the Internet. Instances in the private subnet can establish outbound connections to the Internet via the public subnet using Network Address Translation (NAT).

Creates:

A /16 network with two /24 subnets. Public subnet instances use Elastic IPs to access the Internet. Private subnet instances access the Internet via Network Address Translation (NAT). (Hourly charges for NAT devices apply.)

The diagram illustrates the VPC configuration. It shows an 'Amazon Virtual Private Cloud' containing a 'Public Subnet' and a 'Private Subnet'. The 'Public Subnet' is connected to the 'Internet, S3, DynamoDB, SNS, SQS, etc.' via a 'NAT' gateway. The 'Private Subnet' is connected to the 'Public Subnet' via a 'NAT' gateway.

Step 2: Make a VPC with Public and Private Subnets

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IPv4 CIDR block: (65531 IP addresses available)

IPv6 CIDR block: ☒ No IPv6 CIDR Block
☐ Amazon provided IPv6 CIDR block
☐ IPv6 CIDR block owned by me

VPC name:

Public subnet's IPv4 CIDR: (251 IP addresses available)

Availability Zone:

Public subnet name:

Private subnet's IPv4 CIDR: (251 IP addresses available)

Availability Zone:

Private subnet name:

You can add more subnets after AWS creates the VPC.

Specify the details of your NAT gateway ([NAT gateway rates apply](#)).

Elastic IP Allocation ID:

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Step 3: Verify and Name your Route Tables

☐	Private Rou...	rtb-0c8bf8c2e9926359d	subnet-02e6b25a2250a28de	-	No	vpc-0e4786aebca7f55f3 ...	830190540046
☒	Public Rout...	rtb-0e5e1191ec519f068	subnet-07d1effb7bc827071	-	Yes	vpc-0e4786aebca7f55f3 ...	830190540046

Step 4: Verify your Internet Gateway is Attached

Internet gateways (1/3) Info							Refresh	Actions	Create internet gateway
Filter internet gateways							< 1 > Settings		
☐	Name	Internet gateway ID		State	VPC ID		Owner		
☐	MyVPC_IG	igw-01fe3b25c08a54d6c		✔ Attached	vpc-0f9167e028cfdd9e6 MyVPC		830190540046		
☒	-	igw-0e081cfa20f4bcca		✔ Attached	vpc-0e4786aebca7f55f3 BitBeat 200 ...		830190540046		
☐	-	igw-90a440ea		✔ Attached	vpc-b5e150c8		830190540046		

Problems

I had no issues in this lab.

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